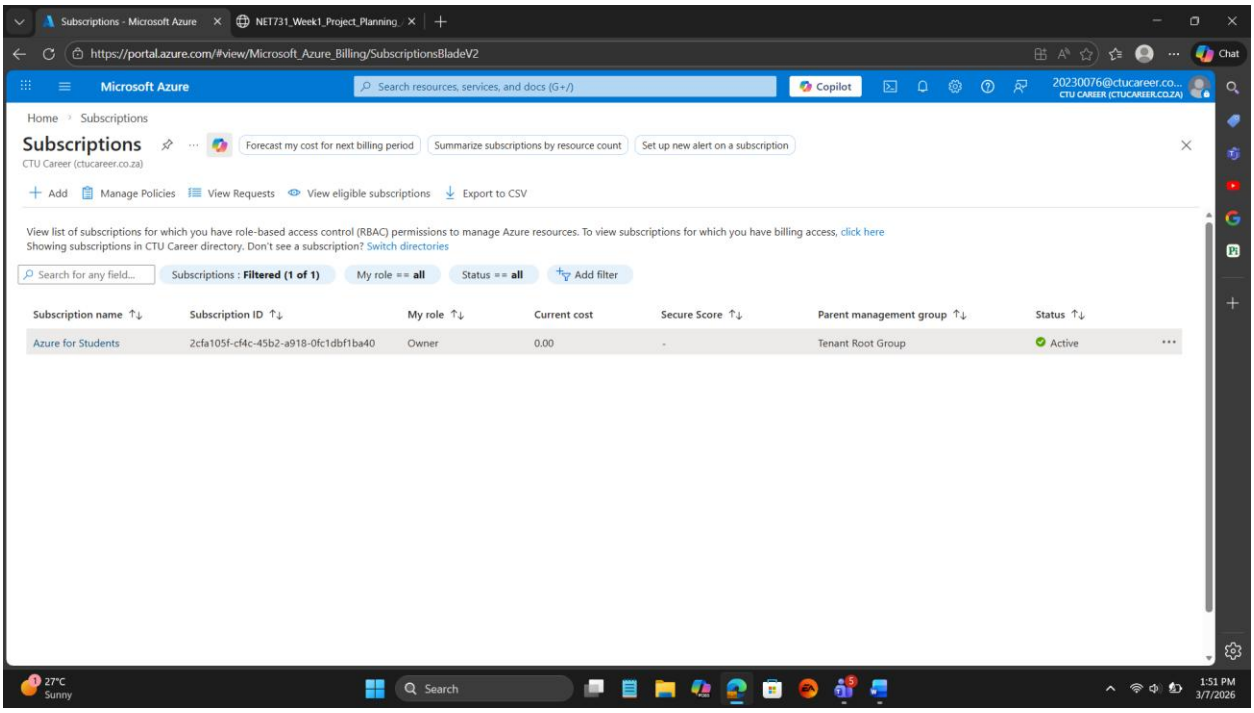
ONNEILE. 20230076. NET731. WEEK 1

TASK 1



TASK 2

1.What type of organization is CTU Training Solutions?(size, industry, purpose)

-CTU Training Solutions is an institution of higher education and skills development.

-CTU is a private tertiary education provider that is registered with the department of higher education and training in South Africa.

-It is a medium sized organization that has an estimate of 200 to 500 employees.

-Its purpose is to upskill people, improve employability, offers diplomas, national qualification, short courses and international certifications in various fields such as information technology, engineering, etc.

2.What are the main IT needs of the organization?(consider: staff, students, data, applications)

- With staff needs, they will requires learning management systems, email platforms, collaboration tools, etc.

-Student needs would be, online learning systems, access to course materials, student portals, collaboration tools, authentication system.

-Data applications, learning management systems, collaboration tools and student information systems

3. Why would a cloud solution (Azure) be more beneficial than on-premises infrastructure for CTU?

-Scalability, cloud resources are able to scale accommodate high volumes of traffic

-Cost efficiency, virtualized resources are cheaper to run, maintain and azure has different subscription options

-Remote/hybrid learning supported as cloud platforms and be accessed anywhwere in the country

4. What security and compliance requirements should be considered?

-Data protection for sensitive information by using access control and encryption

-IAM,MFA and RBAC

-Network security systems like firewalls, DDOS protection

5. What is the expected number of users who will access the cloud environment?

+- 5000 to 8000 full time and part time students across the country(12 campuses)

Academic and admin staff +- 1000

BIBLIOGRAPHY

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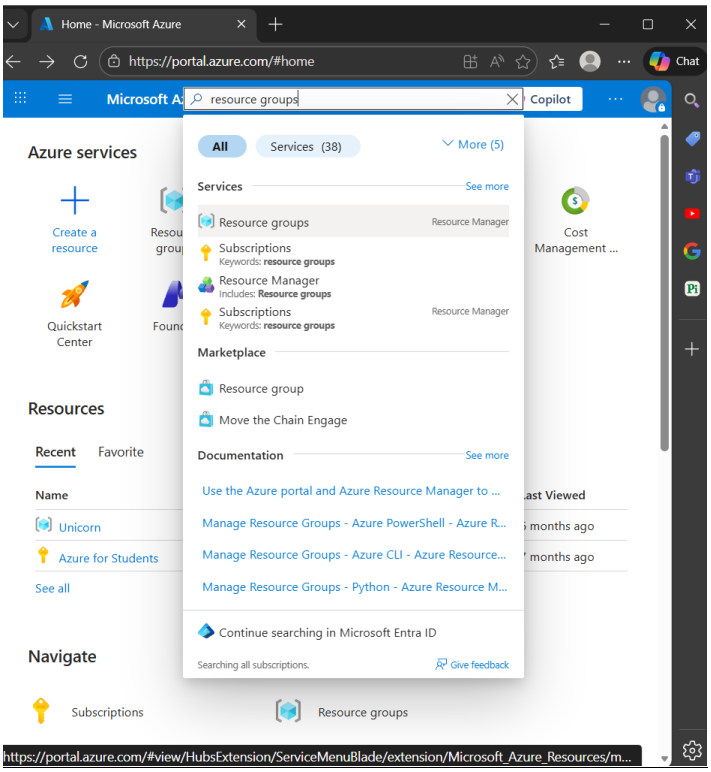
-Protection of Personal Information Act (2013) Protection of Personal Information Act 4 of 2013. https://www.gov.za/documents/protection-personal-information-act

-National Institute of Standards and Technology (2020) Cloud Computing Overview. https://www.nist.gov/itl/cloud

TASK 3

Azure Service	Purpose / Why you need it	Week it is used
Resource Group	It is a container that is used to organize and manage azure resources	01
Virtual Network (VNet)	A private network in Azure	02
Virtual Machine (VM – B1s)	A cloud based computer used to run apps and services	03
Microsoft Entra ID	Manages user identities and secure access to Azure resources	04
Network Security Group (NSG)	Controls network traffic by allowing or blocking access to Azure resources.	05
Azure Storage Account	Stores files, backups, and application data securely in the cloud.	05
Microsoft Defender for Cloud	Provides security monitoring and threat protection for Azure resources	06
Azure Monitor	Tracks performance, logs and health of Azure resources	07
Azure Cost Management	Helps monitor and control cloud spending.	08

TASK 4


Home - Microsoft Azure
https://portal.azure.com/#home
Chat
Microsoft A
resource groups
Copilot
Azure services
Create a resource
Quickstart Center
Cost Management ...
All
Services (38)
More (5)
Services
See more
Resource groups
Resource Manager
Subscriptions
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Resource Manager
Includes: Resource groups
Subscriptions
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Resource Manager
Marketplace
Resource group
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Documentation
See more
Use the Azure portal and Azure Resource Manager to ...
Manage Resource Groups - Azure PowerShell - Azure R...
Manage Resource Groups - Azure CLI - Azure Resource...
Manage Resource Groups - Python - Azure Resource M...
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Give feedback
Resources
Recent
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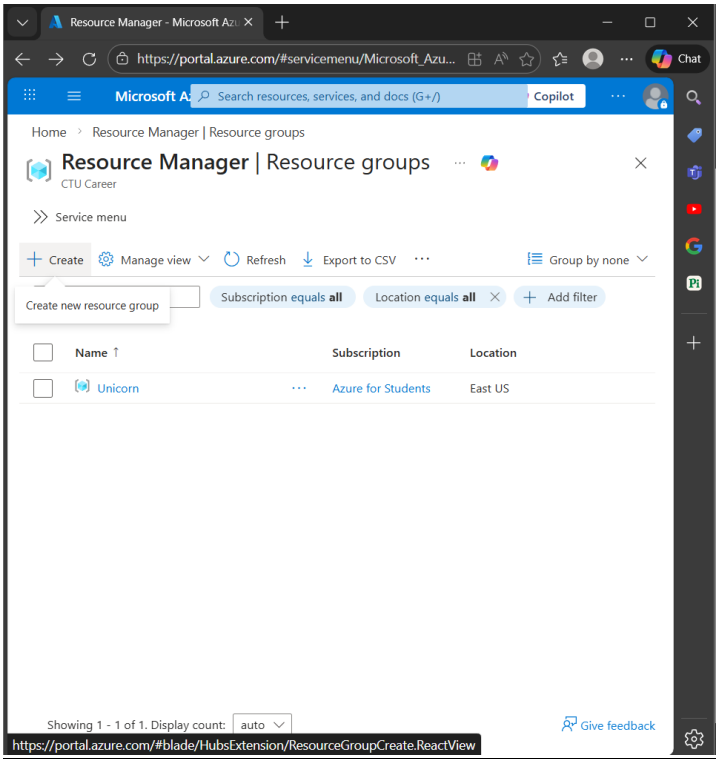
Resource Manager - Microsoft Azu
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Search resources, services, and docs (G+/)
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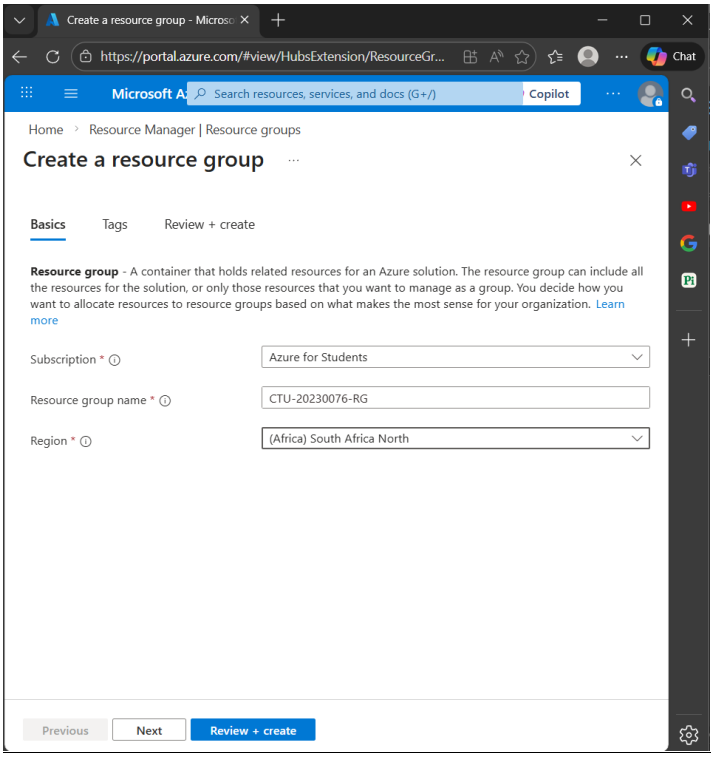
Home
Resource Manager | Resource groups
Resource Manager | Resource groups
CTU Career
Service menu
Create
Manage view
Refresh
Export to CSV
Group by none
Create new resource group
Subscription equals all
Location equals all
Add filter
Name
Subscription
Location
Unicorn
Azure for Students
East US
Showing 1 - 1 of 1. Display count: auto
Give feedback
https://portal.azure.com/#blade/HubsExtension/ResourceGroupCreate.ReactView


Create a resource group - Microso
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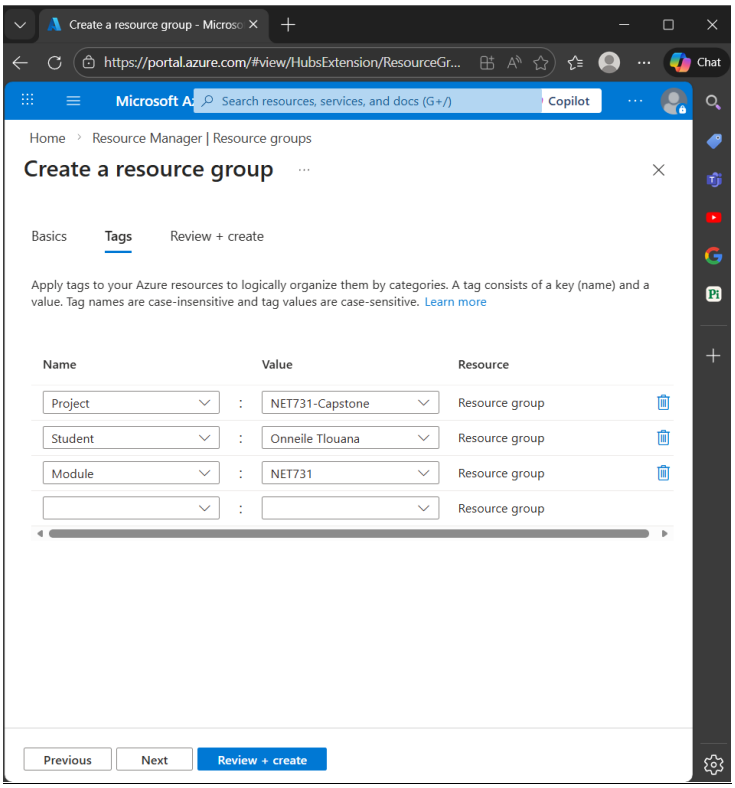
Chat
Microsoft A
Search resources, services, and docs (G+/)
Copilot
Home
Resource Manager | Resource groups
Create a resource group
Basics
Tags
Review + create
Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. Learn more
Subscription *
Azure for Students
Resource group name *
CTU-20230076-RG
Region *
(Africa) South Africa North
Previous
Next
Review + create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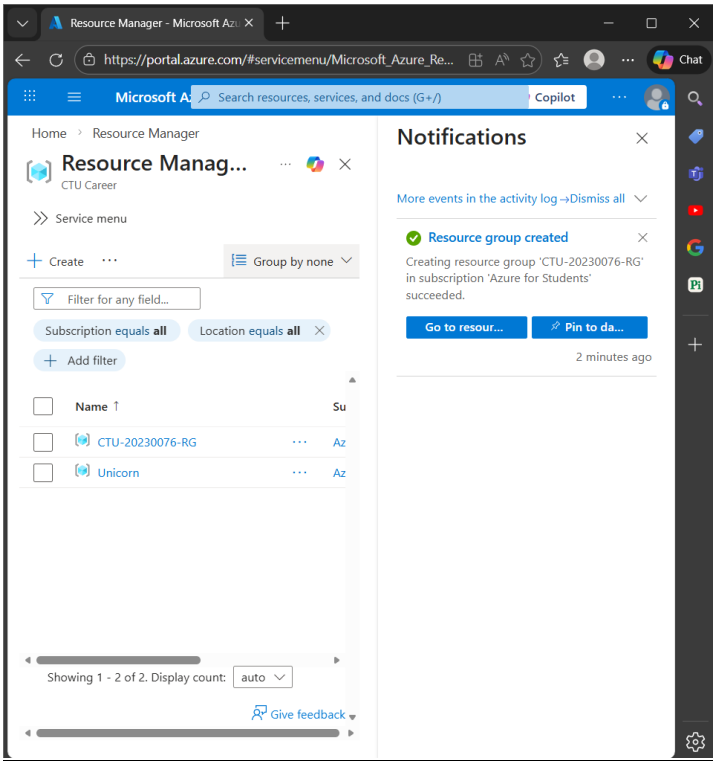
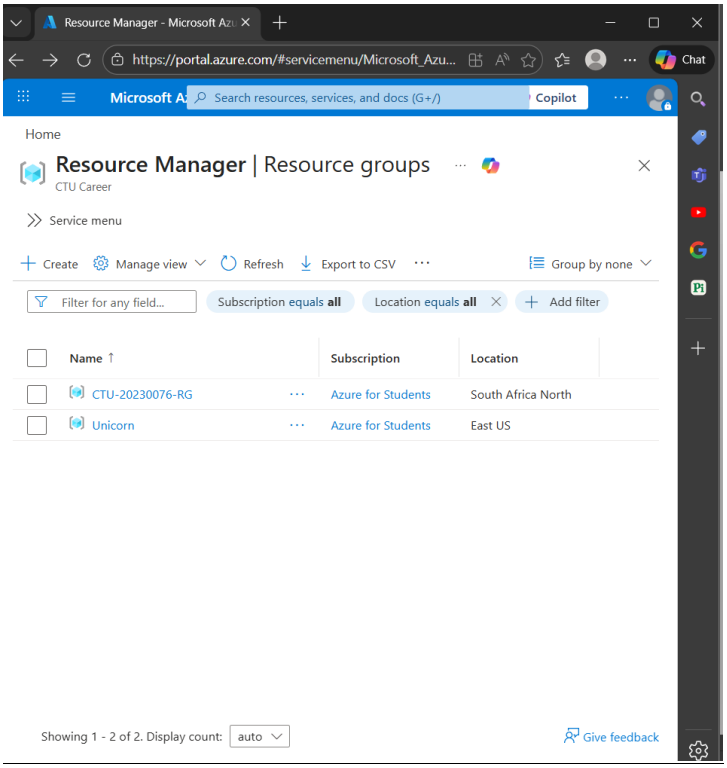
Create a resource group
Basics
Tags
Review + create
Apply tags to your Azure resources to logically organize them by categories. A tag consists of a key (name) and a value. Tag names are case-insensitive and tag values are case-sensitive. Learn more
Name
Value
Resource
Project
NET731-Capstone
Resource group
Student
Onneile Tlouana
Resource group
Module
NET731
Resource group
Resource group
Previous
Next
Review + create


Resource Manager - Microsoft Azu
https://portal.azure.com/#servicemenu/Microsoft_Azure_Re...
Chat
Microsoft A
Search resources, services, and docs (G+/)
Copilot
Home
Resource Manager
Resource Manag...
CTU Career
Service menu
Create
Group by none
Filter for any field...
Subscription equals all
Location equals all
Add filter
Name
CTU-20230076-RG
Unicorn
Showing 1 - 2 of 2. Display count:
auto
Give feedback
Notifications
More events in the activity log →Dismiss all
Resource group created
Creating resource group 'CTU-20230076-RG' in subscription 'Azure for Students' succeeded.
Go to resour...
Pin to da...
2 minutes ago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Create
Manage view
Refresh
Export to CSV
Group by none
Filter for any field...
Subscription equals all
Location equals all
Add filter
Name
Subscription
Location
CTU-20230076-RG
Azure for Students
South Africa North
Unicorn
Azure for Students
East US
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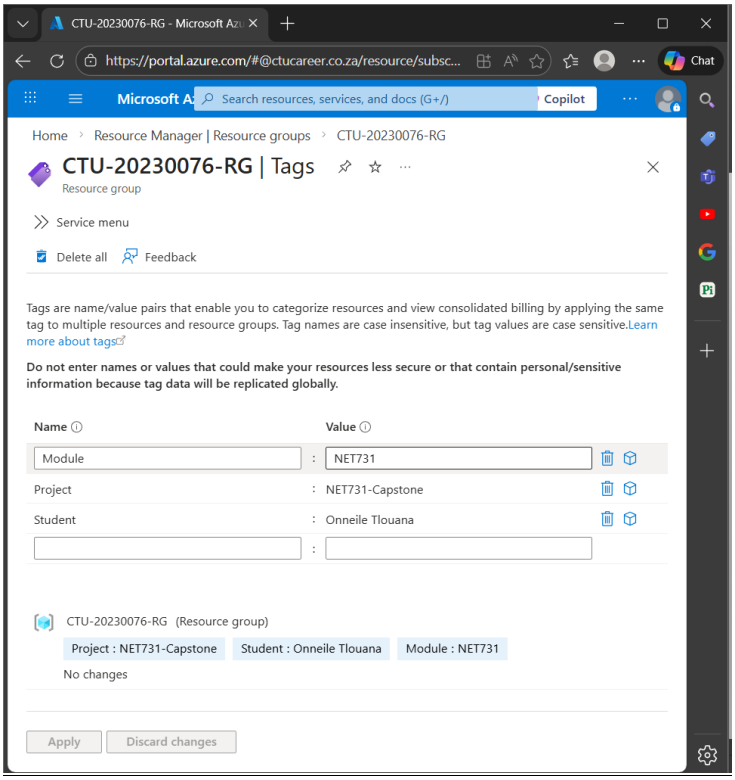
auto
Give feedback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Search resources, services, and docs (G+/)
Copilot
Home
Resource Manager | Resource groups
CTU-20230076-RG
CTU-20230076-RG | Tags
Resource group
Service menu
Delete all
Feedback
Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. Tag names are case insensitive, but tag values are case sensitive.Learn more about tags
Do not enter names or values that could make your resources less secure or that contain personal/sensitive information because tag data will be replicated globally.
Name
Value
Module
NET731
Project
NET731-Capstone
Student
Onneile Tlouana
CTU-20230076-RG (Resource group)
Project : NET731-Capstone
Student : Onneile Tlouana
Module : NET731
No changes
Apply
Discard changes